MSTN (Myostatin) Gene: An overlooked target for gene therapy?

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Aims and Objectives

- Identify the length and complementarity of MSTN nucleotide sequences in *Homo sapiens, Pan troglodytes, Pan paniscus* and *Oryctolagus cuniculus.*
- Explore the potential of genetic modification therapies such as *CRISPR/Cas9* from phylogenetic analysis of the above species.

Role of MSTN

- MSTN, also known as growth differentiation factor 8 (GDF-8) (Joulia-Ekaza D. and Cabello G., 2007), encodes Myostatin - a myokine.
- It acts as a negative regulator of skeletal muscle growth (G-Cadavid N., Taylor W., Yarasheski K. et al., 1998)

Comparison of Muscle Fibres Across Species

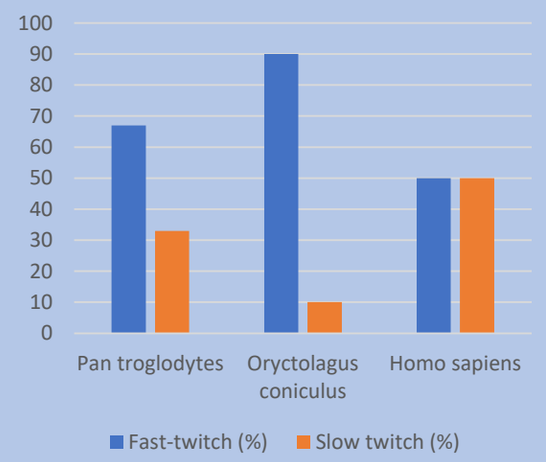

Fig 1. Graph collating data to provide an overview of the ratio of fast-twitch and slow-twitch muscle fibre across *Pan troglodytes, Oryctolagus cuniculus* and *Homo sapiens* (Buss G., 2020; Huang Y., Wenzhao L. 2013; O'Neill et al. 1987)

MSTN Role in Development of Therapies

- MSTN has been shown to control muscle mass by inhibiting cell proliferation and differentiation, for example inhibiting myoblast differentiation into myotubes. (Deng B., Zhang F, Wen J. et al., 2017).

MSTN Role in Development of Therapies (cont.)

- By identifying differences in the sequence of MSTN in the identified species we can begin to develop potential genetic modification therapies to treat muscle-wasting diseases.

Methods

- MSTN .fasta sequences obtained from NCBI and aligned using BLAST.

Species	GenBank Accession No.
Homo sapien	NC_000002.12
Pan troglodyte	NC_036881.1
Pan paniscus	XM_034954676.1
Oryctolagus cuniculus	NC_013675.1

Fig 2. Table displaying the GenBank Accession No. for the MSTN with the corresponding species (Buss G., 2020; NCBI, 2020).

- Bootstrapped (300) Maximum Liklihood and Neighbour-Joining Tree created using MegaX.
- Tamura-Mei (1993) model of evolution was used to construct phylogenetic tree as this distinguishes between transitions, allows different transversions and uses an unequal base frequency. In this way there is a large degree of freedom of which to analyse phylogeny leading to a greater reliability.
- BLAST sequencing of *H. sapiens* and *P. troglodytes* (accession no.: Query_62573) (Zang et al. 2000).

Maximum Likelihood Tree

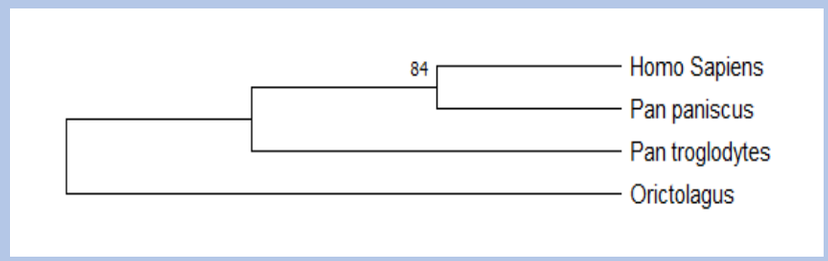

Fig 3. Phylogenetic Maximum Liklihood Tree created using MegaX based on the Tamura-Mei (1993) model of evolution. *Oryctolagus cuniculus* used to root the tree as they are descended from a common ancestor of the *Homo* and *Pan* taxons (Huang Y., Wenzhao L., 2013) *Pan Paniscus* was used as four species are required to form a phylogenetic tree and they are a known related species to both *H. sapiens and P. troglofytes.*

BLAST alignment subject: *Homo sapiens*

Query Species	Total Length	Per. Iden	Accession No.
Orictolagus cuniculus	4912	82.89%	Query_48759
Pan Troglodytes	5333	98.67%	Query_57433

Fig 4. BLAST nucleotide alignment carried out with subject of *Homo sapiens* MSTN gene against queries *Oryctolagus cuniculus* and *Pan troglodytes*. In line with figure 2 BLAST alignment shows a higher percentage identity between *P. troglodytes* and *H. sapiens* than *O. cuniculus* (Zhang et al. 2000)

What does this mean?

- As MSTN gene length decreases, muscle fibre ratio compared to humans increases.
- Therefore, would shortening of the MSTN gene result in increased muscle fibre development in humans?
- Genetic engineering therapies could be developed which combat the effects of muscle wasting diseases such as muscular dystrophy.

Bibliography

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